

| DATE      | TEAM ID         | PROJECT NAME           | VERSION | STATUS   |
|-----------|-----------------|------------------------|---------|----------|
| 1.10.2026 | SWTID-2026-8437 | Power BI Analytics FSD | 1.0     | Approved |

## 1. SYSTEM ARCHITECTURE & OVERVIEW

This FSD specifies functional requirements and DAX technical rules for implementing Power BI Reporting on top of automated ticket classification flows. Row-Level Security (RLS) dynamically filters underlying data models: users with the **EEE role** can query and view tickets where Branch = EEE, non-EEE users are denied access to protected records, and system **Administrators** maintain full visibility.

## 2. FUNCTIONAL REQUIREMENTS BREAKDOWN

| Req ID | Functional Requirement Description                                          | Target Component           |
|--------|-----------------------------------------------------------------------------|----------------------------|
| FR-01  | Import ticket dataset and map Branch field attributes cleanly.              | Power Query ETL            |
| FR-02  | Implement dynamic Row-Level Security (RLS) based on logged-in user.         | Power BI RLS Engine        |
| FR-03  | Restrict non-EEE users from viewing Branch = EEE records.                   | DAX Security Filter        |
| FR-04  | Ensure EEE users have full read access to EEE branch ticket data.           | DAX Security Filter        |
| FR-05  | Ensure Administrators bypass branch filters and view global data.           | Admin Role Assignment      |
| FR-06  | Provide KPI cards for Total Tickets, Open Tickets, and SLA Met %            | Dashboard Visuals          |
| FR-07  | Enable cross-filtering across Category, Priority, and Branch visual charts. | Interactive Report Page    |
| FR-08  | Handle empty or null Branch values securely.                                | Power Query / RLS Fallback |

## 3. DAX FORMULAS & METRICS SPECIFICATION

Key DAX measures designed for ticket classification analytics:

```
// Total Ticket Count Measure
Total Tickets = COUNTROWS(Fact_Tickets)

// EEE Branch Ticket Ratio
EEE Ticket Ratio =
DIVIDE(
    CALCULATE([Total Tickets], Fact_Tickets[Branch] = "EEE"),
    [Total Tickets],
    0
)

// Dynamic RLS DAX Rule (Applied on Dim_User or Fact_Tickets)
[Branch] = LOOKUPVALUE(Dim_User[AssignedBranch], Dim_User[UserEmail], USERPRINCIPALNAME())
|| USERPRINCIPALNAME() IN {"admin@company.com"}
```

4. SECURITY ARCHITECTURE & RLS RULES

| Security Role   | DAX Filter Expression / Logic                  | Permitted Visibility Scope             |
|-----------------|------------------------------------------------|----------------------------------------|
| EEE_Branch_User | [Branch] = "EEE"                               | Only records where Branch = EEE        |
| Non_EEE_User    | [Branch] <> "EEE" && [Branch] = USER_BRANCH( ) | Non-EEE records matching user's branch |
| Administrator   | No DAX Filter Applied (Unrestricted)           | Global dataset across all branches     |

5. RISK & MITIGATION ANALYSIS

| Identified Risk                  | Severity | Mitigation Strategy                                                                                                |
|----------------------------------|----------|--------------------------------------------------------------------------------------------------------------------|
| RLS bypass via export to Excel   | High     | Disable "Analyze in Excel" and restrict detailed underlying data export permissions in Power BI Service workspace. |
| Null or blank Branch field value | Medium   | Enforce default value Unassigned during Power Query ETL step to prevent data leaking.                              |
| Dataset refresh delay            | Low      | Configure scheduled refresh 4x daily using Enterprise On-Premises Data Gateway.                                    |

6. STAKEHOLDER SIGN-OFF

| Role                   | Name  | Date  | Signature |
|------------------------|-------|-------|-----------|
| Lead BI Developer      | _____ | _____ | _____     |
| Solutions Architect    | _____ | _____ | _____     |
| Product Owner / Client | _____ | _____ | _____     |